



NCIE-014

Database & Storage Engineering Specification

**In development — for human review of full expansion —
Version 1.0**

Gives Codex implementation-level persistence design beneath the approved NCIE-001–013 foundation: canonical-to-physical mapping governance, relational/analytical/object/search/vector/graph/time-series persistence design, registry/memory/Evidence persistence, physical keys/indexing/partitioning/replication, caching, classification/residency/encryption enforcement, retention/backup/DR/migration/reconciliation engineering, database security, storage observability/performance/cost, and persistence testing/verification.

Physical-only, upstream-consuming: NCIE-014 owns physical database/storage engineering realization; it does not redefine canonical semantics (NCIE-003), institutional governance (NCIE-008), security/IAM policy (NCIE-009), Evidence/provenance meaning (NCIE-010), API contracts (NCIE-011), UX patterns (NCIE-012) or domain business logic (NCIE-013).

Expands approved NCIE-014 Skeletal v0.1, itself traced to approved NCIE-001 v1.3 through approved NCIE-013 v1.0.

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Maintained by: National Communications Intelligence Ecosystem (NCIE)

Document Governance

0.1 Source-Confirmed Title and Authority Chain

The approved NCIE-014 Skeletal v0.1 Document Governance block records a source-confirmed title 'NCIE-014 — Database & Storage Engineering Specification' and a source-confirmed suite purpose 'Gives Codex implementation-level persistence design.' This matches the NCIE Documentation Suite Master Controlled Register v2.0, Section 3, NCIE-014 entry exactly (same title wording, same Primary Purpose framing), so no title-reconciliation discrepancy exists and no correction is required — unlike the NCIE-013 v1.0 pre-approval correction, this skeleton already carries a source-confirmed title and no separate reconciliation step is needed.

Authority chain: NCIE-001 requirements → NCIE-002 architecture → NCIE-003 canonical data → NCIE-004 engineering baseline → NCIE-005 ARGUS behavioral authority → NCIE-006 Memory & Context architecture → NCIE-007 AI/Agent/Model Orchestration → NCIE-008 Governance, Authority, Accountability & RACI → NCIE-009 Security, IAM & Zero-Trust Architecture → NCIE-010 Evidence, Provenance & Audit Architecture → NCIE-011 API, Integration & Data Connector Specification → NCIE-012 UX/UI & Design System Specification → NCIE-013 Domain Functional Modules Specification → NCIE-014 Database & Storage Engineering Specification. NCIE-014 implements physical persistence for governed canonical semantics; it creates no canonical meaning, institutional authority, security control, Evidence semantics, API contract, UX pattern or domain-business rule of its own.

0.2 Token-Conservation Convention

Every chapter opens with a short Upstream block citing the specific upstream NCIE-001–013 authority the chapter implements, stating only the physical-persistence relationship rather than restating upstream canonical, governance, security, Evidence, integration, UX or domain-business content.

0.3 Governing Database & Storage Invariants Register

These 21 invariants bind every chapter of this document and are stated once here rather than re-derived repeatedly. They are fixed by the approved NCIE-014 Skeletal v0.1 and by the approved NCIE-001–013 authority chain; no institutional decision is required to adopt them. The upstream canonical, governance, security, Evidence/provenance, integration, UX and domain-module invariants already binding from NCIE-003 and NCIE-008–013 remain fully in force and are cross-referenced, not restated, in the chapters below.

- PHYSICAL SCHEMA ≠ CANONICAL SEMANTIC MODEL.
- DATABASE RECORD ≠ INSTITUTIONAL TRUTH.
- REPLICATION ≠ AUTHORITATIVE CURRENT STATE.
- INDEX ≠ SYSTEM OF RECORD.
- SEARCH INDEX ≠ CANONICAL STORE.
- VECTOR INDEX ≠ MEMORY AUTHORITY.
- GRAPH PROJECTION ≠ CANONICAL OWNERSHIP.
- CACHE ≠ DURABLE RECORD.
- BACKUP ≠ CURRENT AUTHORIZATION.
- RESTORE ≠ REAUTHORIZATION.
- RETENTION ≠ LEGAL HOLD RELEASE.
- DELETION ≠ HISTORICAL ERASURE WHERE GOVERNED HISTORY MUST REMAIN.
- ENCRYPTION ≠ AUTHORIZATION.
- DATABASE ADMINISTRATION ≠ CONTENT AUTHORITY.
- PARTITIONING ≠ SECURITY/TENANT BOUNDARY BY ITSELF.
- DATA LOCALITY ≠ RESIDENCY COMPLIANCE BY ITSELF.
- SCHEMA MIGRATION ≠ SEMANTIC REDEFINITION.
- MIGRATION SUCCESS ≠ RECONCILIATION COMPLETE.
- PERFORMANCE OPTIMIZATION ≠ PERMISSION TO DROP PROVENANCE.
- ARCHIVAL ≠ INVALIDATION.
- IMMUTABILITY MECHANISM ≠ TRUTH OF CONTENT.

0.4 Persistence/Upstream Boundary Note

NCIE-014 owns physical database/storage engineering realization. It consumes NCIE-003 canonical semantics, NCIE-004 technology-family constraints, NCIE-006 Memory/Context roles, NCIE-008 institutional decision rights, NCIE-009 security/IAM controls, NCIE-010 Evidence/provenance/audit semantics, NCIE-011 API/integration contracts, NCIE-012 UX/data-presentation needs and NCIE-013 domain-module workloads, without redefining any of them. Where a matter requires a technology selection, a numeric threshold (RPO/RTO, retention period, partition/replica count, cache TTL) or a residency/sovereignty rule not fixed by an approved source, this document states the physical-persistence framework and defers the choice to Human Review rather than inventing an answer. Approval of the NCIE-014 Skeletal v0.1 authorized this full expansion only; it did not itself approve any technology selection, numeric threshold, retention rule or ownership assignment proposed below.

0.5 Human Review Focus Classification

Every chapter's Human Review Focus item carries one of seven labels: RESOLVED BY UPSTREAM BASELINE; PROPOSED DESIGN DEFAULT; INSTITUTIONAL CONFIRMATION REQUIRED; SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED; LEGAL/POLICY CONFIRMATION REQUIRED; ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED; or SOURCE/WORKFLOW DISCOVERY REQUIRED — each additionally marked Blocking or Non-Blocking, and each stating precisely what it blocks. Per directive item 62, a Chapter 33 register item that restates an identical NCIE-008–013 institutional, security, Evidence, integration, UX or domain-business question cross-references that upstream Human Review ID instead of duplicating it; a new NCIE-014 item is created only for a distinct persistence-engineering decision. Chapter 33 consolidates every open item, extracted directly from the chapters so the register cannot drift from its sources (mechanically verified per §33.4).

0.6 Status Convention

Skeletal-structure approval authorizes expansion; it does not approve any individual technology selection, numeric threshold, retention rule or ownership assignment proposed below — those remain IN DEVELOPMENT / FOR HUMAN REVIEW until a separate approval act. Nothing in this document may be cited as an approved institutional decision, an implemented store, a validated migration/recovery procedure, or a completed test.

0.7 Salient Things to Note Discipline

Directive item 75 requires this expansion to programmatically verify an explicit '<N>.0 Salient Things to Note' subsection in all 34 chapters, continuing the discipline established after the NCIE-009 Chapter-27 omission (found by independent human examination and corrected) and reinforced in every full expansion since. Every chapter below renders its Salient Things to Note subsection as an unconditional part of the same build-time helper (NCIE-014's lib.py raises a build error if a chapter's salient content is missing), and Chapter 34's mechanical audit (§34.2) re-verifies this against the final document.

0.8 Document Amendment History

Version: v0.1

Change: Approved skeletal review edition (structural/conceptual review only; full expansion not yet authorized).

Version: v1.0

Change: Full expansion of the approved NCIE-014 Skeletal v0.1 into the complete 34-chapter specification per the NCIE-014 v0.1→v1.0 Claude Full-Expansion Production Directive: canonical-to-physical mapping governance, six consolidated matrices, technology/numeric-invention and current-state/historical-integrity audits, Human Review Register (Chapter 33) and full traceability/handover (Chapter 34). Formal title source-confirmed at skeletal stage; no reconciliation discrepancy found (§0.1).

Authority: Approved NCIE-001 Master PRD v1.3 through approved NCIE-013 Domain Functional Modules Specification v1.0, and the approved NCIE-014 Skeletal v0.1.

Scope: Controlled full expansion only — the approved 34-chapter skeletal structure, chapter purposes and boundaries are preserved exactly; no chapter was added, removed, merged, split or renumbered.

Status: v1.0 IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION. The approved v0.1 skeleton remains historically preserved as the expansion architecture.

Document	NCIE-014 Database & Storage Engineering Specification
Edition	In development — Version 1.0 (Full Expansion of Approved Skeletal v0.1)
Requirements authority	NCIE-001 Master Product Requirements Document, Version 1.3 — APPROVED
Architecture authority	NCIE-002 System Architecture & Technical Design, Version 0.5 — APPROVED
Canonical data authority	NCIE-003 Data Model & Data Dictionary, Version 0.4 — APPROVED
Engineering baseline	NCIE-004 Technology Stack & Engineering Blueprint, Version 1.2 — APPROVED
ARGUS behavioral authority	NCIE-005 ARGUS Intelligence Assistant Specification, Version 1.3 — APPROVED
Memory & Context authority	NCIE-006 Memory & Context Architecture Specification, Version 1.2 — APPROVED
Orchestration authority	NCIE-007 AI, Agent & Model Orchestration Specification, Version 1.2 — APPROVED
Governance authority	NCIE-008 Governance, Authority, Accountability & RACI Specification, Version 1.0 — APPROVED
Security authority	NCIE-009 Security, IAM & Zero-Trust Architecture, Version 1.0 — APPROVED
Evidence/provenance/audit authority	NCIE-010 Evidence, Provenance & Audit Architecture, Version 1.0 — APPROVED
Integration authority	NCIE-011 API, Integration & Data Connector Specification, Version 1.0 — APPROVED
UX/UI authority	NCIE-012 UX/UI & Design System Specification, Version 1.0 — APPROVED
Domain-module authority	NCIE-013 Functional Modules Specification, Version 1.0 — APPROVED
Prior baseline	NCIE-014 Database & Storage Engineering Specification, Skeletal v0.1 — APPROVED FOR FULL EXPANSION (historically preserved)
Status	IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION until explicitly approved; the approved v0.1 skeleton remains historically preserved as the expansion architecture

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Mandate, Authority Chain & Persistence Boundary

Establish NCIE-014 ownership of physical database/storage engineering within the approved NCIE suite.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: Production-Ready register; NCIE-003 Ch.32; NCIE-004; NCIE-006/009/010/011.

1.0 Salient Things to Note

NCIE-014 implements persistence for approved semantics; it cannot redefine canonical entities, governance, IAM, Evidence, APIs or domain rules — Table 1 fixes this boundary and Table 2 maps every object class to its physical role without creating semantic ownership.

1.1 Mandate and Scope

NCIE-014 owns physical persistence engineering: database/storage realization, physical schemas, indexes, partitioning, retention implementation, backup/restore persistence, and storage-specific performance/reconciliation. It consumes NCIE-003 canonical semantics and must not redefine them.

1.2 Persistence Boundary — What NCIE-014 Does Not Own

NCIE-014 does not own canonical data semantics, institutional governance, security/IAM policy, Evidence/provenance meaning, API contracts, UX patterns or domain business logic. It owns only the physical realization layer beneath all of them.

Table 1. NCIE-014 persistence boundary.

Concern	Owned By
Product requirements/scope	NCIE-001
System architecture	NCIE-002
Canonical data model	NCIE-003
Engineering baseline/technology families	NCIE-004
ARGUS behavioral authority	NCIE-005
Memory & Context architecture	NCIE-006
AI/Agent/Model orchestration	NCIE-007
Institutional governance, RACI	NCIE-008
Security, IAM, Zero-Trust enforcement	NCIE-009
Evidence lifecycle, provenance, audit	NCIE-010
API/event/connector/integration contracts	NCIE-011
UX/UI, presentation, interaction, Design System	NCIE-012
Domain business rules, workflows, analytical products	NCIE-013
Physical database/storage engineering	NCIE-014 (this document)
Deployment/operations, detailed testing	NCIE-015/016 (downstream)

1.3 Physical-Store Allocation Matrix (directive item 55)

The consolidated matrix below maps canonical/logical object classes to their physical persistence roles, authority source, security class, history requirement and unresolved technology choice, without creating semantic ownership — semantic authority remains NCIE-003 regardless of which physical store realizes an object class.

Table 2. Physical-Store Allocation Matrix.

Object Class	Physical Role	Authority Source	Security Class	History Requirement	Technology Status
Canonical transactional entities	Relational/transactional store (Ch.4)	NCIE-003	Per NCIE-009 classification	Temporal fields required	Open — HR14-4-1
Analytical aggregates/projections	Analytical/columnar store (Ch.5)	NCIE-003 (via lineage)	Per source classification	Lineage retained, not canonical	Open — HR14-5-1
Files/documents/large artifacts	Object/blob store (Ch.6)	NCIE-003/NCIE-010	Per object classification	Version/integrity retained	Open — HR14-6-1
Search-indexed content	Search index (Ch.7)	NCIE-003 (source of truth elsewhere)	Authorization-filtered	Non-canonical, rebuildable	Open — HR14-7-1
Embeddings/semantic vectors	Vector index (Ch.8)	NCIE-006 (source of truth elsewhere)	Authorization-filtered	Non-canonical, rebuildable	Open — HR14-8-1
Relationship/graph projections	Graph store (Ch.9)	NCIE-003 (source of truth elsewhere)	Per source classification	Reconciled, not canonical	Open — HR14-9-1
Telemetry/KPI time-series	Time-series store (Ch.10)	NCIE-003/NCIE-013	Per source classification	Rollup/retention tiers apply	Open — HR14-10-1
Registry/VPF/Memory/Content state	Registry & Memory stores (Ch.11-12)	NCIE-006/007	Per NCIE-009 classification	Version/expiry per upstream policy	Open — HR14-11-1
Evidence/provenance/audit records	Evidence/audit store (Ch.13)	NCIE-010	Heightened assurance	Immutable/historically preserved	Open — HR14-13-1

1.R RACI / Control Ownership

Table 3. Chapter 1 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Adopt/confirm NCIE-014 persistence architecture	Accountable Authority Class (NCIE-008, to be assigned)	Persistence Architecture Owner (to be assigned)	NCIE-003/004/009/010 chapter owners	All institutional role holders

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR14-1-1: Confirm NCIE-014 acceptance/sign-off authority (skeletal Preliminary Human Review Theme 1).

Traceability: Production-Ready register; NCIE-003 Ch.32; NCIE-004; NCIE-006/009/010/011.

Persistence Engineering Doctrine

Define common persistence principles binding every physical store this document specifies.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-002 layered architecture; NCIE-003; NCIE-009/010.

2.0 Salient Things to Note

Physical optimization must preserve semantics, provenance, authorization boundaries and historical reconstructability — durability, consistency, traceability, least privilege and recoverability are stated explicitly per store, never assumed from a vendor default.

2.1 Durability, Consistency and Isolation

Every physical store must state its durability guarantee, consistency model and isolation behavior explicitly rather than by assumption; a store's default configuration is never treated as an approved guarantee until Human Review confirms it.

2.2 Traceability, Least Privilege and Recoverability

Physical design choices must preserve traceability (who/what/when for every write), least-privilege access at the storage layer, and recoverability to a known-good state — optimization that removes any of these three properties is out of scope for this document without an explicit upstream governance decision.

2.3 Portability and Explicit Authority

Where feasible, physical design avoids unnecessary vendor lock-in (portability), and every design choice cites its explicit upstream authority (NCIE-002/003/004/009/010) rather than an implicit assumption about what 'good practice' requires.

2.R RACI / Control Ownership

Table 4. Chapter 2 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Approve persistence principles and unresolved trade-offs	Accountable Authority Class (to be assigned)	Persistence Doctrine Owner (to be assigned)	NCIE-002/003/009/010 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR14-2-1: Approve persistence principles and unresolved trade-offs (durability/consistency/isolation defaults by store class).

Traceability: NCIE-002 layered architecture; NCIE-003; NCIE-009/010.

Canonical-to-Physical Mapping Governance

Define controlled mapping from NCIE-003 canonical models to physical structures.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 canonical dictionary/ERDs.

3.0 Salient Things to Note

PHYSICAL SCHEMA ≠ CANONICAL SEMANTIC MODEL; denormalization/projection cannot silently change meaning — Table 5 fixes the mapping-register discipline linking every physical structure back to its NCIE-003 canonical source and version.

3.1 Mapping Register

Every physical structure that realizes a canonical entity, field, type, constraint or derived field is recorded in a mapping register linking it back to its NCIE-003 source and version — denormalization, projection or physical renaming never silently changes the canonical meaning it represents.

3.2 Physical Naming and Version Linkage

Physical names may diverge from canonical names for engineering reasons (reserved words, length limits, convention), but the mapping register preserves the link explicitly; every mapped structure carries a version reference so a canonical model change can be traced to every physical structure it affects.

3.3 Canonical-to-Physical Mapping Matrix (directive item 56)

The consolidated matrix below maps canonical entities/fields/identifiers/relationships to physical structures and migration/version references; all semantic authority remains NCIE-003 regardless of physical representation.

Table 5. Canonical-to-Physical Mapping Matrix.

Canonical Element (NCIE-003)	Physical Structure Class	Mapping Type	Version Linkage	Semantic Authority
Canonical entity	Table/collection/node (per store class)	1:1 or 1:N physical realization	Mapping-register entry versioned with schema migration	NCIE-003
Canonical field	Column/attribute/property	Direct or derived (computed) field	Field-level version reference	NCIE-003
Canonical identifier	Primary/foreign key or equivalent	Direct mapping, no re-derivation of meaning	Identifier scheme versioned per NCIE-003 Ch. 15	NCIE-003
Canonical relationship	Foreign key, join table, or graph edge	Direct or projected (Ch.9)	Relationship-version reference	NCIE-003
Canonical type/constraint	Physical type/check constraint	Direct or engineering-equivalent type	Type-mapping table versioned	NCIE-003
Derived/computed field	Materialized or virtual column	Explicitly marked derived, never presented as canonical source data	Derivation-rule version reference	NCIE-003 (rule basis) + deriving chapter

3.R RACI / Control Ownership

Table 6. Chapter 3 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm physical naming/mapping conventions	Accountable Authority Class (to be assigned)	Mapping Governance Owner (to be assigned)	NCIE-003 data-dictionary owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR14-3-1: Confirm physical naming/mapping conventions.

Traceability: NCIE-003 canonical dictionary/ERDs.

Relational / Transactional Persistence Design

Define physical persistence for transactional/canonical workloads.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 entities; NCIE-004 technology families.

4.0 Salient Things to Note

Database rows are implementations of canonical objects, not independent semantic authorities — denormalization is only permitted where explicitly recorded in the Chapter 3 mapping register.

4.1 Tables, Keys and Constraints

Transactional tables realize canonical entities per the Chapter 3 mapping register; keys and constraints enforce the physical integrity rules defined in Chapter 15, and never introduce a business rule the canonical model does not already require.

4.2 Transactions, Concurrency and Temporal Fields

Transaction boundaries and concurrency-control strategy (optimistic/pessimistic) are stated per workload; temporal fields (created/updated/effective/superseded) are mandatory on every canonical-entity table so historical reconstruction (Chapter 27) remains possible.

4.3 Normalization/Denormalization Choices

Denormalization for performance is permitted only where the mapping register (Chapter 3) records the derived relationship explicitly; database rows are implementations of canonical objects, not independent semantic authorities.

4.R RACI / Control Ownership

Table 7. Chapter 4 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm relational technology and transaction requirements	Accountable Authority Class (to be assigned)	Transactional Store Owner (to be assigned)	NCIE-003/004 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR14-4-1: Confirm selected relational technology and transaction requirements (no product mandated by this document per directive item 8).

Traceability: NCIE-003 entities; NCIE-004 technology families.

Analytical / Columnar Persistence Design

Define analytical-store physical design.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 analytical layer; NCIE-003 analytics; NCIE-013 modules.

5.0 Salient Things to Note

Analytical projections/aggregates must retain lineage and cannot become canonical truth — every materialization cites its refresh cadence and canonical/domain source.

5.1 Facts, Dimensions and Columnar Layout

Analytical structures (facts/dimensions, columnar layout) are physical projections of canonical and domain-module data; every fact/dimension table cites its canonical/domain source in the mapping register.

5.2 Aggregates, Materialization, Refresh and Lineage

Aggregates and materializations carry an explicit refresh cadence and lineage reference; historical partitions preserve prior aggregate states so a later reconciliation can explain a discrepancy rather than silently overwrite history.

5.R RACI / Control Ownership

Table 8. Chapter 5 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm analytical-store technology and workload priorities	Accountable Authority Class (to be assigned)	Analytical Store Owner (to be assigned)	NCIE-002/003/013 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR14-5-1: Confirm analytical-store technology and workload priorities.

Traceability: NCIE-002 analytical layer; NCIE-003 analytics; NCIE-013 modules.

Object / Blob Storage Design

Define storage of files, documents, large artifacts and immutable payloads.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Evidence/source objects; NCIE-010; NCIE-009.

6.0 Salient Things to Note

Object availability does not establish Evidence status; metadata, classification and integrity remain governed by NCIE-010/NCIE-009 regardless of physical storage tier.

6.1 Object Identity, Metadata and Versioning

Every stored object carries an identity, metadata (classification, source, integrity hash) and version history; object availability does not establish Evidence status — that comes only through the NCIE-010 lifecycle.

6.2 Lifecycle and Access

Object lifecycle (creation, versioning, tiering, deletion) and access controls follow NCIE-009 authorization and NCIE-010 custody requirements; large artifacts are chunked/streamed per engineering convention without altering the object's registered integrity metadata.

6.R RACI / Control Ownership

Table 9. Chapter 6 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm object-store technology and sensitive-object classes	Accountable Authority Class (to be assigned)	Object Store Owner (to be assigned)	NCIE-009/010 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR14-6-1: Confirm object-store technology and sensitive-object classes.

Traceability: NCIE-003 Evidence/source objects; NCIE-010; NCIE-009.

Search Index Engineering

Define physical search indexing.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-006 search; NCIE-010 discovery; NCIE-009 authorization.

7.0 Salient Things to Note

SEARCH INDEX ≠ CANONICAL STORE; stale/index-only data must not be represented as authoritative current state — every result is authorization-filtered before it reaches a requester.

7.1 Index Documents and Source Reference

Every indexed document carries a reference back to its canonical/source record and a freshness timestamp; the search index itself is never the authoritative store — SEARCH INDEX ≠ CANONICAL STORE.

7.2 Updates, Deletion and Authorization Filtering

Index updates/tombstones propagate from source changes on an explicit cadence; the authorization-filtering interface ensures a search result never surfaces content the requester's NCIE-009 authorization would not permit reading directly.

7.R RACI / Control Ownership

Table 10. Chapter 7 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm search technology and freshness expectations	Accountable Authority Class (to be assigned)	Search Index Owner (to be assigned)	NCIE-006/009/010 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR14-7-1: Confirm search technology and freshness expectations.

Traceability: NCIE-006 search; NCIE-010 discovery; NCIE-009 authorization.

Vector Index / Embedding Persistence

Define vector/embedding persistence for semantic retrieval.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-006 vector/search; NCIE-009; NCIE-010.

8.0 Salient Things to Note

VECTOR INDEX ≠ MEMORY AUTHORITY; similarity does not grant disclosure authority or Evidence status — namespace scoping and authorization filters apply identically to search retrieval.

8.1 Embedding Reference, Source and Version

Every stored embedding references its source content, source version and generating model version; VECTOR INDEX ≠ MEMORY AUTHORITY — similarity results never grant disclosure authority or Evidence status on their own.

8.2 Namespace, Freshness and Re-Embedding

Vector namespaces are scoped to prevent cross-boundary leakage; deletion/re-embedding follows source-content lifecycle changes, and authorization filters apply to retrieval identically to Chapter 7's search-index filtering.

8.R RACI / Control Ownership

Table 11. Chapter 8 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm vector technology, embedding lifecycle and sensitive-data policy	Accountable Authority Class (to be assigned)	Vector Store Owner (to be assigned)	NCIE-006/009/010 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR14-8-1: Confirm vector technology, embedding lifecycle and sensitive-data policy.

Traceability: NCIE-006 vector/search; NCIE-009; NCIE-010.

Graph Persistence & Relationship Projections

Define physical graph/relationship storage.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 graph/relationships; NCIE-010 provenance graph.

9.0 Salient Things to Note

GRAPH PROJECTION ≠ CANONICAL OWNERSHIP and LINEAGE/RELATIONSHIP ≠ CAUSALITY — every projection is rebuildable from canonical sources rather than being an independent authority.

9.1 Nodes, Edges and Canonical References

Graph nodes/edges reference canonical entities/relationships (Chapter 3 mapping) rather than inventing independent identity; GRAPH PROJECTION ≠ CANONICAL OWNERSHIP.

9.2 Temporal Relationships, Provenance and Reconciliation

Relationships carry temporal validity and provenance; a graph rebuild/reconciliation process restores the projection from canonical sources without asserting new relationships the canonical/source data does not support. LINEAGE/RELATIONSHIP ≠ CAUSALITY.

9.R RACI / Control Ownership

Table 12. Chapter 9 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm graph technology and projection ownership	Accountable Authority Class (to be assigned)	Graph Store Owner (to be assigned)	NCIE-003/010 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Non-Blocking) HR14-9-1: Confirm graph technology and projection ownership; does not block the reconciliation/rebuild discipline already defined in the interim.

Traceability: NCIE-003 graph/relationships; NCIE-010 provenance graph.

Time-Series & Metrics Persistence

Define storage for KPI/telemetry/time-series workloads.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 temporal/KPI; NCIE-004 observability; NCIE-013 KPI modules.

10.0 Salient Things to Note

Operational telemetry and domain KPI history remain semantically distinct; retention/aggregation must preserve required interpretation — late/corrected data is appended with provenance, never silently overwritten.

10.1 Series Identity, Timestamps and Dimensions

Every time-series carries a stable series identity, timestamp precision and labeled dimensions; operational telemetry (Chapter 29) and domain KPI history (NCIE-013) remain semantically distinct series even where they share a storage technology.

10.2 Rollups, Late Data and Provenance

Rollup/retention tiers are stated per series class; late-arriving or corrected data is appended with provenance rather than silently overwriting a prior aggregate, so a later query can distinguish original from corrected values.

10.R RACI / Control Ownership

Table 13. Chapter 10 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm time-series technology and retention/rollup requirements	Accountable Authority Class (to be assigned)	Time-Series Store Owner (to be assigned)	NCIE-004/013 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR14-10-1: Confirm time-series technology and retention/rollup requirements.

Traceability: NCIE-003 temporal/KPI; NCIE-004 observability; NCIE-013 KPI modules.

Agent, Model, Tool & VPF Registry Persistence

Define physical persistence for orchestration registries and Validation Artifacts.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-003/004/007; NCIE-009 security.

11.0 Salient Things to Note

Persistence cannot create Agent registration, model eligibility, Tool authority or VPF approval — registry state remains governed upstream regardless of which physical store realizes it.

11.1 Registry Structures and History

Agent Pattern/Definition, Model Registry, Tool Registry and VPF Profile records are persisted with full version/history; persistence cannot create Agent registration, model eligibility, Tool authority or VPF approval — registry state remains governed entirely upstream (NCIE-007).

11.2 Validation Artifact References

Validation Artifacts are stored with references to the registry entries and governance events that produced them, preserving the NCIE-007 approval chain rather than re-deriving it from storage state.

11.R RACI / Control Ownership

Table 14. Chapter 11 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm physical store choices and history requirements	Accountable Authority Class (to be assigned)	Registry Store Owner (to be assigned)	NCIE-007/009 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [LEGAL/POLICY CONFIRMATION REQUIRED] (Blocking) HR14-11-1: Confirm physical store choices and history requirements for Agent/Model/Tool/VPF registries.

Traceability: NCIE-003/004/007; NCIE-009 security.

Memory & Context Persistence

Define physical realization of NCIE-006 Memory/Context storage roles.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-006 v1.2; NCIE-009.

12.0 Salient Things to Note

Stored Memory ≠ current truth/authorization; Agent Context remains bounded and ephemeral where required — the durable/ephemeral cache boundary is explicit, never inferred from storage behavior.

12.1 User/Room/Investigation Memory and Preferences

Stored Memory is a persistence realization of NCIE-006 roles exactly — it is never current truth or authorization on its own; expiry, invalidation and provenance follow NCIE-006 policy precisely, not a storage-layer default.

12.2 Agent Context and Cache Boundaries

Agent Context remains bounded and ephemeral where NCIE-006 requires it; the cache boundary between durable Memory and ephemeral Context is explicit so a downstream reader cannot mistake cached Context for durable Memory.

12.R RACI / Control Ownership

Table 15. Chapter 12 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm store allocation, retention and ephemeral-persistence policy	Accountable Authority Class (to be assigned)	Memory Store Owner (to be assigned)	NCIE-006/009 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR14-12-1: Confirm store allocation, retention and ephemeral-persistence policy for Memory/Context.

Traceability: NCIE-006 v1.2; NCIE-009.

Evidence, Provenance & Audit Persistence

Define physical storage for Evidence, custody, provenance and audit.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-010 v1.0; NCIE-009.

13.0 Salient Things to Note

Physical immutability/tamper evidence supports integrity but does not establish truth; audit/provenance remain distinct streams even when physically co-located.

13.1 Evidence Versions, Custody and Provenance

Evidence versions, custody events and the provenance graph are persisted per NCIE-010 lifecycle requirements; physical immutability/tamper-evidence mechanisms support integrity but do not themselves establish truth of content — IMMUTABILITY MECHANISM ≠ TRUTH OF CONTENT.

13.2 Audit Events and Integrity Metadata

Audit events remain a distinct stream from provenance/custody events even where co-located physically; integrity metadata (hashes, signatures) is retained alongside historical state so a later verification can confirm nothing was altered outside the governed custody chain.

13.R RACI / Control Ownership

Table 16. Chapter 13 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm storage classes and heightened assurance requirements	Accountable Authority Class (to be assigned)	Evidence Store Owner (to be assigned)	NCIE-009/010 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR14-13-1: Confirm storage classes and heightened assurance requirements for Evidence/provenance/audit persistence.

Traceability: NCIE-010 v1.0; NCIE-009.

Security, Identity, Secrets & Key-Metadata Persistence Interface

Define persistence boundaries for security-related records without becoming the secrets/key authority.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-009 Ch.3-12.

14.0 Salient Things to Note

Secrets/key material must follow NCIE-009; database administration does not create content/security authority — this chapter persists references and metadata only, never secret material.

14.1 Identity, Authorization and Session Data

Identity references, authz-policy data and session metadata are persisted per NCIE-009 requirements; database administration over these stores does not create content or security authority — DATABASE ADMINISTRATION ≠ CONTENT AUTHORITY.

14.2 Secret and Key-Metadata References

This document persists only references/metadata for secrets and keys, never secret material itself, following NCIE-009 policy exactly; security events are persisted with privileged separation from ordinary administrative access.

14.R RACI / Control Ownership

Table 17. Chapter 14 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm which security records may reside in which stores	Accountable Authority Class (to be assigned)	Security Persistence Owner (to be assigned)	NCIE-009 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR14-14-1: Confirm which security records may reside in which stores.

Traceability: NCIE-009 Ch.3-12.

Physical Keys, Constraints & Referential Integrity

Define physical integrity constraints.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-003 identifiers/cardinality.

15.0 Salient Things to Note

Physical constraints enforce implementation consistency but cannot replace governance/business validation — temporal uniqueness and soft-delete rules keep historical rows queryable.

15.1 Keys, Constraints and Nullability

Primary/unique/foreign keys, check constraints and nullability rules are derived directly from NCIE-003 identifiers and cardinality; physical constraints enforce implementation consistency but cannot replace governance/business validation performed upstream.

15.2 Temporal Uniqueness and Soft-Delete Rules

Temporal uniqueness (unique within an effective-date window) and soft-delete/tombstone rules are stated explicitly so historical rows remain queryable rather than being physically removed where governed history must remain (Chapter 22).

15.R RACI / Control Ownership

Table 18. Chapter 15 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm constraint strictness for ingestion/staging versus canonical stores	Accountable Authority Class (to be assigned)	Physical Integrity Owner (to be assigned)	NCIE-003 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Non-Blocking) HR14-15-1: Confirm constraint strictness for ingestion/staging versus canonical stores; does not block the baseline constraint discipline already defined in the interim.

Traceability: NCIE-003 identifiers/cardinality.

Indexing & Query Access Paths

Define indexes/access paths for required workloads.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-001 workloads; NCIE-003; NCIE-013.

16.0 Salient Things to Note

PERFORMANCE OPTIMIZATION ≠ PERMISSION TO DROP PROVENANCE or weaken authorization filtering — every index is justified by a named, source-supported query pattern.

16.1 Primary, Secondary and Composite Indexes

Indexes are derived from source-supported workload patterns identified in NCIE-001/NCIE-013 rather than speculative optimization; covering-index and composite-path choices are recorded with the query pattern that justifies them.

16.2 Optimization Boundary

PERFORMANCE OPTIMIZATION ≠ PERMISSION TO DROP PROVENANCE — index maintenance and query rewriting may never bypass the authorization-filtering interfaces defined in Chapters 7-8 or remove fields required for provenance/audit reconstruction.

16.R RACI / Control Ownership

Table 19. Chapter 16 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm critical query/workload priorities	Accountable Authority Class (to be assigned)	Indexing Owner (to be assigned)	NCIE-001/013 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Non-Blocking) HR14-16-1: Confirm critical query/workload priorities; does not block the source-supported indexing discipline already defined in the interim.

Traceability: NCIE-001 workloads; NCIE-003; NCIE-013.

Partitioning, Sharding & Distribution

Define physical distribution strategy.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 scale; NCIE-009 residency; NCIE-003 temporal/domain models.

17.0 Salient Things to Note

PARTITIONING ≠ SECURITY BOUNDARY BY ITSELF; distribution must preserve residency, authorization and reconciliation — no partition/shard count is invented by this document.

17.1 Partition Keys and Sharding Concept

Partition keys (tenant/domain/time) are chosen from NCIE-003 temporal/domain models; PARTITIONING ≠ SECURITY/TENANT BOUNDARY BY ITSELF — a partition boundary is never presented as an authorization boundary without an explicit NCIE-009 control alongside it.

17.2 Rebalance, Hotspots and Cross-Partition Behavior

Rebalance operations, hotspot mitigation and cross-partition query behavior are stated with their provenance/sovereignty implications; no partition count or rebalance threshold is invented by this document (directive item 9).

17.R RACI / Control Ownership

Table 20. Chapter 17 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm partition/shard criteria and sovereignty constraints	Accountable Authority Class (to be assigned)	Distribution Owner (to be assigned)	NCIE-002/003/009 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR14-17-1: Confirm partition/shard criteria and sovereignty constraints.

Traceability: NCIE-002 scale; NCIE-009 residency; NCIE-003 temporal/domain models.

Replication, Consistency & Authoritative-Read Semantics

Define replication and consistency behavior.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-006 current-state; NCIE-009 authz; NCIE-011 APIs.

18.0 Salient Things to Note

REPLICA ≠ AUTHORITATIVE CURRENT STATE. Workflows requiring current authority/state must use an approved authoritative-read path — Table 21 fixes which workload classes require which read path.

18.1 Primary/Replica Roles and Consistency Levels

Primary/replica roles, consistency-level concepts and failover behavior are stated per store; REPLICA ≠ AUTHORITATIVE CURRENT STATE — a replica read is only ever a candidate answer, never an authoritative one, until routed through an approved authoritative-read path.

18.2 Stale-Read Indicators and Reconciliation

Every replica-served response carries a stale-read indicator (lag/timestamp) so a consuming workflow can decide whether the staleness is acceptable; reconciliation on rejoin resolves divergence without silently discarding either side's history.

18.3 Authoritative-Read Matrix (directive item 57)

The consolidated matrix below identifies workloads that require authoritative current state versus permitted stale/replica/cache reads, with explicit staleness indicators and reconciliation paths.

Table 21. Authoritative-Read Matrix.

Workload Class	Read Requirement	Staleness Indicator	Reconciliation Path
Authorization/entitlement checks (NCIE-009)	Authoritative current state required	N/A — must route to primary/authoritative path	N/A
Evidence/custody state reads (NCIE-010)	Authoritative current state required	N/A — authoritative path only	N/A
Domain analytics/dashboards (NCIE-013)	Stale/replica reads permitted	Replica lag timestamp surfaced in UX (NCIE-012)	Periodic replica-primary reconciliation (Ch.27)
Search/vector retrieval (Ch.7-8)	Stale/index reads permitted	Index freshness timestamp surfaced	Re-index on source change
Cached UI reads (Ch.19)	Stale/cache reads permitted within TTL	Cache-age indicator surfaced where sensitive	Invalidation on source write
Case/Investigation state changes (NCIE-013 Ch.9)	Authoritative current state required	N/A — authoritative path only	N/A

18.R RACI / Control Ownership

Table 22. Chapter 18 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm consistency requirements by workload	Accountable Authority Class (to be assigned)	Replication Owner (to be assigned)	NCIE-006/009/011 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR14-18-1: Confirm consistency requirements by workload (acceptable replica lag, authoritative-read routing).

Traceability: NCIE-006 current-state; NCIE-009 authz; NCIE-011 APIs.

Caching & Materialized Views

Define caching/materialization boundaries.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-006 Context/cache; NCIE-009; NCIE-011.

19.0 Salient Things to Note

CACHE ≠ DURABLE RECORD; cached authorization, Evidence status or decision state cannot silently override current authority — sensitive classes get stricter TTL or exclusion pending Human Review.

19.1 Cache Keys, TTL and Invalidation

Cache keys, TTL concept and invalidation triggers are stated per cacheable data class; CACHE ≠ DURABLE RECORD — cached authorization, Evidence status or decision state can never silently override current authority.

19.2 Materialized Views and Sensitive-Cache Controls

Materialized views carry refresh/staleness indicators identical in spirit to Chapter 18's authoritative-read discipline; sensitive data classes (per NCIE-009) receive stricter TTL or are excluded from caching entirely, pending Human Review confirmation.

19.R RACI / Control Ownership

Table 23. Chapter 19 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm cacheable data classes and invalidation requirements	Accountable Authority Class (to be assigned)	Cache Owner (to be assigned)	NCIE-006/009/011 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR14-19-1: Confirm cacheable data classes and invalidation requirements.

Traceability: NCIE-006 Context/cache; NCIE-009; NCIE-011.

Data Classification, Residency & Sovereignty Enforcement

Translate governance/security classifications into physical placement controls.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-008/009; NCIE-003 classification.

20.0 Salient Things to Note

DATA LOCALITY ≠ RESIDENCY COMPLIANCE BY ITSELF; metadata, backups, replicas and indexes must be included in placement analysis — Table 24 covers every copy type, not just the primary store.

20.1 Classification Mapping and Placement

Every physical copy of classified data (primary, replica, index, cache, backup, archive) is mapped to its NCIE-008/009 classification and permitted region/zone; DATA LOCALITY ≠ RESIDENCY COMPLIANCE BY ITSELF — a single primary-store location claim is never sufficient.

20.2 Cross-Border Controls

Cross-border replication, backup or index placement requires an explicit sovereignty-approved control; unresolved locations/policies remain open Human Review items rather than assumed defaults.

20.3 Residency-Copy Matrix (directive item 58)

The consolidated matrix below maps primary, replica, index, cache, backup, DR and archive copies to their classification/residency/sovereignty requirements.

Table 24. Residency-Copy Matrix.

Copy Type	Classification Basis	Residency Requirement	Sovereignty Status
Primary store	NCIE-008/009 classification per object class	Per approved region/zone	Open — HR14-20-1
Replica (Ch.18)	Inherits primary classification	Must match or be explicitly approved cross-region	Open — HR14-20-1
Search/vector index (Ch.7-8)	Inherits source classification	Per approved region/zone	Open — HR14-20-1
Cache (Ch.19)	Inherits source classification	Per approved region/zone; stricter for sensitive classes	Open — HR14-20-1
Backup (Ch.23)	Inherits primary classification	Per approved region/zone; encryption required	Open — HR14-20-1
DR copy (Ch.24)	Inherits primary classification	Per approved cross-site policy	Open — HR14-20-1
Archive (Ch.22)	Inherits primary classification, retention-adjusted	Per approved region/zone	Open — HR14-20-1

20.R RACI / Control Ownership

Table 25. Chapter 20 RACI / control ownership.

Function	Accountable (NCIE-	Responsible	Consulted	Informed
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	008)	(Persistence)		
Confirm approved residency/sovereignty rules and locations	Accountable Authority Class (to be assigned)	Residency Enforcement Owner (to be assigned)	NCIE-008/009 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR14-20-1: Confirm approved residency/sovereignty rules and locations covering primary stores, replicas, indexes, caches, backups and archives (Table 24).
- Traceability: NCIE-008/009; NCIE-003 classification.*

Encryption, Tokenization & Protected-Field Storage

Define physical protection of sensitive fields/data.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 protected identity; NCIE-009 crypto/privacy.

21.0 Salient Things to Note

ENCRYPTION ≠ AUTHORIZATION; tokenization/masking must preserve governed linkage and minimum-necessary access — no encryption algorithm is mandated by this document.

21.1 At-Rest Encryption and Field Protection

At-rest encryption properties, field-level protection and key-separation interfaces follow NCIE-009 crypto policy; ENCRYPTION ≠ AUTHORIZATION — an encrypted field still requires NCIE-009 authorization to decrypt and read, never encryption status alone.

21.2 Tokenization, Masking and Protected Identity

Token references and masking preserve governed linkage back to the protected identity (NCIE-003) with minimum-necessary access; no encryption algorithm is mandated by this document (directive item 41 — key-separation interface only).

21.R RACI / Control Ownership

Table 26. Chapter 21 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm protection policy by data class	Accountable Authority Class (to be assigned)	Encryption/Tokenization Owner (to be assigned)	NCIE-003/009 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR14-21-1: Confirm protection policy by data class.

Traceability: NCIE-003 protected identity; NCIE-009 crypto/privacy.

Retention, Legal Hold, Archival & Disposition Engineering

Implement logical retention/disposition physically.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-006 retention; NCIE-008 governance; NCIE-010 Ch.24.

22.0 Salient Things to Note

RETENTION ≠ LEGAL HOLD RELEASE; disposition must preserve required history/tombstones and not falsify Evidence/audit state — no retention period is invented by this document.

22.1 Retention Class, Hold and Archive Tier

Retention class, legal-hold flag and archive tier are physical implementations of governance decisions (NCIE-008/NCIE-010); RETENTION ≠ LEGAL HOLD RELEASE — expiry of a retention period never physically deletes a record still under an active legal hold.

22.2 Tombstones, Purge Verification and Metadata Retention

Deletion workflows use tombstones where governed history must remain (DELETION ≠ HISTORICAL ERASURE); purge verification confirms disposal completed as governed, and disposal never falsifies Evidence/audit state by silently removing the record of that the disposal occurred.

22.3 Retention/Disposition Matrix (directive item 59)

The consolidated matrix below maps logical retention/hold/archive/disposition states to physical implementation obligations without inventing periods.

Table 27. Retention/Disposition Matrix.

Logical State	Physical Obligation	Tombstone Required	Period Source
Active retention	Full accessibility in primary/replica stores	No	Open — HR14-22-1
Legal hold	Deletion/archival blocked regardless of retention-period expiry	N/A — hold overrides	NCIE-008 hold authority
Archive tier	Moved to lower-cost tier per Ch.31; metadata retained in primary index	No	Open — HR14-22-1
Disposition-eligible	Deletion workflow triggered per governed schedule	Yes, where governed history must remain	Open — HR14-22-1
Post-disposal	Record purged; tombstone/metadata retained per governance	Yes	NCIE-008/010 disposal policy

22.R RACI / Control Ownership

Table 28. Chapter 22 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm retention periods, hold authority and	Accountable Authority Class (to be assigned)	Retention/Disposition Owner (to be assigned)	NCIE-006/008/010 owners	Assurance/Audit (NCIE-008)

disposal rules

Human Review Focus

- [LEGAL/POLICY CONFIRMATION REQUIRED] (Blocking) HR14-22-1: Confirm retention periods, hold authority and disposal rules (Table 27; no period invented by this document).

Traceability: NCIE-006 retention; NCIE-008 governance; NCIE-010 Ch.24.

Backup, Restore & Point-in-Time Recovery

Define physical backup/recovery architecture.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004 resilience; NCIE-009/010 recovery.

23.0 Salient Things to Note

BACKUP ≠ CURRENT AUTHORIZATION and RESTORE ≠ REAUTHORIZATION; restored state must reconcile current identity/eligibility/policy — Table 29 fixes the reconciliation path for every recovery scenario, including migration and repair.

23.1 Backup Sets, Snapshots and PITR

Backup sets, snapshot concept and point-in-time recovery (PITR) capability are stated per store class with catalog/isolation/encryption requirements; BACKUP ≠ CURRENT AUTHORIZATION.

23.2 Restore Validation and Recovery Sequence

RESTORE ≠ REAUTHORIZATION — a restored copy must reconcile current identity, eligibility and policy state before being treated as authoritative; restore validation and recovery sequence are stated explicitly rather than assumed.

23.3 Recovery/Reconciliation Matrix (directive item 60)

The consolidated matrix below maps failure/restore/DR/migration scenarios to restore source, current-state reconciliation, authorization/eligibility refresh, integrity checks and acceptance evidence.

Table 29. Recovery/Reconciliation Matrix.

Scenario	Restore Source	Current-State Reconciliation	Authorization/Eligibility Refresh	Integrity Check	Acceptance Evidence
Single-store restore (Ch.23)	Latest valid backup/snapshot	Reconcile against authoritative-read path (Ch.18) before serving	Required before restored copy is authoritative	Checksum/catalog verification	Restore-validation record
Cross-site DR failover (Ch.24)	DR replica/copy	Reconcile post-failback per Ch.24 fencing/reconciliation	Required at promotion and at failback	Split-brain/consistency check	DR-drill or failover acceptance record
Schema migration rollback (Ch.25)	Pre-migration backup or reverse migration	Reconcile schema version and semantic compatibility	N/A unless identity schema affected	Migration-log/checksum verification	Rollback-validation record
Legacy data migration exception (Ch.26)	Source legacy system extract	Reconcile via Ch.26 exception/reconciliation workflow	N/A unless identity records affected	Row-count/checksum reconciliation	Cutover acceptance record
Reconciliation/repair (Ch.27)	Divergent/duplicate state within governed stores	Reconcile per approved repair proposal	N/A unless authorization records affected	Post-repair verification	Repair-approval and audit record

23.R RACI / Control Ownership

Table 30. Chapter 23 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm RPO/RTO and backup classes	Accountable Authority Class (to be assigned)	Backup/Recovery Owner (to be assigned)	NCIE-004/009/010 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR14-23-1: Confirm RPO/RTO and backup classes (Table 29; no numeric value invented by this document).

Traceability: NCIE-004 resilience; NCIE-009/010 recovery.

Disaster Recovery & Cross-Site Persistence

Define cross-site persistence/failover.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-004 DR; NCIE-009; NCIE-010.

24.0 Salient Things to Note

Failover must not activate stale policy, secrets, Agent eligibility or Evidence state; recovery gaps remain explicit rather than hidden by a successful-looking failover.

24.1 Replication, Promotion and Fencing

Cross-site replication/failover, promotion and fencing procedures prevent split-brain conditions; a failover must not activate stale policy, secrets, Agent eligibility or Evidence state — each is explicitly re-validated at promotion.

24.2 Reconciliation and Failback

Post-incident reconciliation and failback restore single-source-of-truth operation without silently discarding writes accepted during the split; recovery gaps (data accepted at one site but not yet replicated) remain explicit rather than hidden.

24.R RACI / Control Ownership

Table 31. Chapter 24 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm DR topology and sovereignty constraints	Accountable Authority Class (to be assigned)	DR Owner (to be assigned)	NCIE-004/009/010 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR14-24-1: Confirm DR topology and sovereignty constraints.

Traceability: NCIE-004 DR; NCIE-009; NCIE-010.

Schema Migration & Database Change Management

Define safe physical schema evolution.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 schema governance; NCIE-004 CI/CD; NCIE-011 contracts.

25.0 Salient Things to Note

SCHEMA MIGRATION ≠ SEMANTIC REDEFINITION; breaking semantic change requires upstream canonical governance — expand/contract preserves compatibility windows and migration provenance.

25.1 Migration Versioning and Expand/Contract

Schema migrations follow an expand/contract pattern with explicit compatibility windows; SCHEMA MIGRATION ≠ SEMANTIC REDEFINITION — a breaking semantic change requires upstream NCIE-003 canonical governance approval before any physical migration proceeds.

25.2 Backfill, Rollback and Availability Risk

Backfill jobs, rollback procedures and availability/lock risk are stated per migration; provenance of the migration (who approved, what changed, when) is retained regardless of migration outcome.

25.R RACI / Control Ownership

Table 32. Chapter 25 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm migration approval and downtime tolerance	Accountable Authority Class (to be assigned)	Migration Owner (to be assigned)	NCIE-003/004/011 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR14-25-1: Confirm migration approval and downtime tolerance.

Traceability: NCIE-003 schema governance; NCIE-004 CI/CD; NCIE-011 contracts.

Data Migration, Backfill & Legacy Reconciliation

Define movement of legacy/historical data into governed stores.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.34; NCIE-010 provenance.

26.0 Salient Things to Note

MIGRATION SUCCESS ≠ RECONCILIATION COMPLETE; counts/checksums alone do not prove semantic correctness — every migrated record retains provenance back to its legacy source.

26.1 Source Inventory, Mapping and Transform

Legacy source inventory, field mapping and transformation rules are recorded before any migration executes; MIGRATION SUCCESS ≠ RECONCILIATION COMPLETE — counts/checksums alone do not prove semantic correctness of the migrated data.

26.2 Validation, Exceptions, Reconciliation and Cutover

Migration validation produces an exception list requiring explicit disposition; provenance of every migrated record traces back to its legacy source, and cutover to the new system of record happens only after reconciliation authority signs off.

26.R RACI / Control Ownership

Table 33. Chapter 26 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm legacy sources, acceptance criteria and reconciliation authority	Accountable Authority Class (to be assigned)	Legacy Migration Owner (to be assigned)	NCIE-003/010 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR14-26-1: Confirm legacy sources, acceptance criteria and reconciliation authority.

Traceability: NCIE-003 Ch.34; NCIE-010 provenance.

Data Lifecycle Reconciliation & Repair

Define detection/repair of missing, duplicate, conflicting or divergent persisted state.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-003/010/011.

27.0 Salient Things to Note

Repair must preserve history and provenance; automated repair cannot silently rewrite institutional state — every repair is proposed, approved, audited and post-verified.

27.1 Reconciliation Jobs and Divergence Detection

Scheduled reconciliation jobs detect duplicate, missing or divergent state across primary/replica/index/cache copies; detection alone never triggers an automated write — repair requires the approval path defined below.

27.2 Repair Proposal, Approval and Post-Repair Verification

Repair must preserve history and provenance; automated repair cannot silently rewrite institutional state — every repair is proposed, approved by the accountable authority, audited, and independently verified after application.

27.R RACI / Control Ownership

Table 34. Chapter 27 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm auto-repair boundaries and approval	Accountable Authority Class (to be assigned)	Reconciliation Owner (to be assigned)	NCIE-003/010/011 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR14-27-1: Confirm auto-repair boundaries and approval.

Traceability: NCIE-003/010/011.

Database Security & Privileged Administration

Define database/storage security administration under NCIE-009.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-009 Ch.8-9; NCIE-008 SoD.

28.0 Salient Things to Note

DATABASE ADMINISTRATION ≠ CONTENT AUTHORITY; privileged maintenance must remain attributable and least-privileged — separation is maintained between operational access and content authority.

28.1 DB Identities, Privileged Roles and Elevation

DB identities, privileged roles and elevation/break-glass procedures follow NCIE-009 identity and NCIE-008 separation-of-duties requirements; DATABASE ADMINISTRATION ≠ CONTENT AUTHORITY — a DBA's ability to access storage never confers authority over the content's meaning or use.

28.2 Row/Field Security Interface and Admin Audit

Row/field-security interfaces enforce NCIE-009 authorization at the storage layer where supported; every privileged administrative action is audited and attributable, with separation maintained between operational DBA access and content/security decision authority.

28.R RACI / Control Ownership

Table 35. Chapter 28 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm DBA privilege model and emergency authority	Accountable Authority Class (to be assigned)	Database Security Owner (to be assigned)	NCIE-008/009 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR14-28-1: Confirm DBA privilege model and emergency authority.

Traceability: NCIE-009 Ch.8-9; NCIE-008 SoD.

Storage Observability, Capacity & Health

Define operational storage observability without preempting NCIE-015.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004 observability; NCIE-010 distinctions; NCIE-015 downstream.

29.0 Salient Things to Note

Telemetry supports operation but is not audit/provenance; capacity signals must not be mistaken for business/data-quality truth — detailed operational execution is handed to NCIE-015.

29.1 Capacity, Latency and Health Signals

Capacity, latency, error rate, saturation, replication lag, backup health and index health are observability signals only; telemetry supports operation but is not audit/provenance — capacity signals must not be mistaken for business or data-quality truth.

29.2 Storage Growth and Alerting Handover

Storage growth trends and alerting thresholds are specified as requirements for NCIE-015 operational implementation, not implemented or executed by this document; this chapter does not preempt NCIE-015 deployment/operations detail.

29.R RACI / Control Ownership

Table 36. Chapter 29 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm storage SLO/capacity thresholds	Accountable Authority Class (to be assigned)	Storage Observability Owner (to be assigned)	NCIE-004/010/015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR14-29-1: Confirm storage SLO/capacity thresholds (no numeric value invented by this document).

Traceability: NCIE-004 observability; NCIE-010 distinctions; NCIE-015 downstream.

Performance Engineering & Query Optimization

Define performance validation/optimization principles.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 NFRs; NCIE-013 workloads.

30.0 Salient Things to Note

Optimization must not weaken correctness, authorization, provenance or historical integrity — no specific performance target is invented by this document.

30.1 Query Plans, Workload Profiles and Indexing

Query-plan analysis, workload profiling and indexing/partition-pruning strategy follow NCIE-002 non-functional requirements and NCIE-013 workload priorities; optimization must not weaken correctness, authorization, provenance or historical integrity.

30.2 Batch/Stream Balance and Regression Detection

Batch/stream processing balance and performance-regression detection are stated as principles; no specific performance target is invented by this document absent an approved source (directive item 9).

30.R RACI / Control Ownership

Table 37. Chapter 30 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm critical workloads and performance targets	Accountable Authority Class (to be assigned)	Performance Engineering Owner (to be assigned)	NCIE-002/013 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Non-Blocking) HR14-30-1: Confirm critical workloads and performance targets; does not block the correctness-first optimization principles already defined in the interim.

Traceability: NCIE-002 NFRs; NCIE-013 workloads.

NCIE-014 — Chapter 31 Expansion

Cost, Tiering & Storage Efficiency

Define governed storage-efficiency architecture.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004 technology/cost; NCIE-008 governance.

31.0 Salient Things to Note

Cost optimization cannot silently shorten retention, weaken redundancy or move data across sovereignty boundaries — any tiering decision touching those areas requires the same Human Review confirmation as the underlying decision.

31.1 Hot/Warm/Cold Tiering and Compression

Hot/warm/cold tiering, compression and deduplication concepts are stated as engineering levers; cost optimization cannot silently shorten retention, weaken redundancy or move data across sovereignty boundaries established in Chapter 20.

31.2 Capacity Planning and Cost Attribution

Capacity planning and cost attribution are stated per store class; any tiering decision that would affect retention, redundancy or residency requires the same Human Review confirmation as the underlying Chapter 20/22/24 decision, not a unilateral cost-driven change.

31.R RACI / Control Ownership

Table 38. Chapter 31 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm cost objectives and permitted tiering	Accountable Authority Class (to be assigned)	Storage Efficiency Owner (to be assigned)	NCIE-004/008 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR14-31-1: Confirm cost objectives and permitted tiering.

Traceability: NCIE-004 technology/cost; NCIE-008 governance.

Database / Storage Testing & Verification

Define persistence test classes without duplicating NCIE-016.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-016 downstream; NCIE-009/010 assurance.

32.0 Salient Things to Note

TEST SPECIFIED ≠ TEST EXECUTED; successful CRUD does not prove recovery, migration, residency, integrity or performance correctness — all nine test classes above are SPECIFIED only.

32.1 Test Class Catalogue

Nine persistence test classes are specified: schema/constraint, migration, backup/restore, DR, reconciliation, security, residency, performance and corruption tests. Each test class states its objective and acceptance basis; TEST SPECIFIED ≠ TEST EXECUTED.

Table 39. Database/Storage Test-Class Catalogue.

Test Class	Objective	Acceptance Basis	Status
Schema/constraint	Verify physical constraints match Chapter 15 integrity rules	Constraint-violation test suite	SPECIFIED
Migration	Verify expand/contract migration compatibility (Ch.25)	Pre/post-migration schema diff and rollback test	SPECIFIED
Backup/restore	Verify backup sets restore to a valid, queryable state (Ch.23)	Restore-validation checklist	SPECIFIED
DR	Verify cross-site failover/failback without split-brain (Ch.24)	DR-drill scenario results	SPECIFIED
Reconciliation	Verify repair workflow detects and resolves divergence without data loss (Ch.27)	Post-repair verification record	SPECIFIED
Security	Verify privileged access/row-field security enforcement (Ch.28)	Access-control test suite	SPECIFIED
Residency	Verify every copy type lands in its approved region/zone (Ch.20)	Placement-audit scan	SPECIFIED
Performance	Verify query/workload behavior under approved targets (Ch.30)	Load/regression test suite	SPECIFIED
Corruption	Verify integrity-check mechanisms detect corrupted/tampered data	Checksum/integrity-verification suite	SPECIFIED

32.2 Test-Status Integrity (directive item 63)

This chapter distinguishes TEST SPECIFIED, IMPLEMENTATION EVIDENCE AVAILABLE, TEST EXECUTED, TEST PASSED, RECOVERY/MIGRATION VALIDATED and HUMAN/PRODUCTION ACCEPTANCE as six distinct states; every test class above is SPECIFIED only — none is claimed executed, passed, recovery/migration-validated or accepted by this document.

32.R RACI / Control Ownership

Table 40. Chapter 32 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Persistence)	Consulted	Informed
Confirm test acceptance authority and required scenarios	Accountable Authority Class (to be assigned)	Persistence Test Owner (to be assigned)	NCIE-009/010/016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR14-32-1: Confirm test acceptance authority and required scenarios.

Traceability: NCIE-016 downstream; NCIE-009/010 assurance.

Human Review Register & Open Persistence Decisions

Consolidate every unresolved persistence-engineering decision exposed across Chapters 1-32 and Chapter 34 into one register: deduplicated, consistently classified, each stating precisely what it blocks. No item here manufactures closure of a decision the approved upstream chain, the approved NCIE-008–013 governance, security, Evidence, integration, UX and domain-business layers, or the approved NCIE-014 Skeletal v0.1 does not itself settle.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-008–013 Human Review registers, plus persistence-engineering-specific unresolved items from Chapters 1-32.

33.0 Salient Things to Note

Do not invent technology selections, RPO/RTO, retention periods or residency rules; deduplicate upstream questions and state precise blocking scope.

33.1 Purpose

This register exists so that no unresolved persistence-engineering decision is silently settled by expansion prose, and so that no chapter's own Human Review Focus item is left outside the register it is meant to feed — a defect found and corrected in the companion NCIE-008 v1.0 document's initial build. Every item from Chapters 1-32 below was extracted programmatically from that chapter's own Human Review Focus section (§33.4). Chapter 33's own item (HR14-33-1) and Chapter 34's own item (HR14-34-1) are each defined once as a fixed value and rendered identically both here and in their own chapter's Human Review Focus bullet, so neither can be omitted from the register or drift from the bullet that also states it.

33.2 Summary

Table 41. Human Review Register summary (mechanically computed).

Classification	Count
ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED	9
INSTITUTIONAL CONFIRMATION REQUIRED	12
LEGAL/POLICY CONFIRMATION REQUIRED	2
RESOLVED BY UPSTREAM BASELINE	1
SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED	10
Total	34
— Blocking	29
— Non-Blocking	5

33.3 Complete Register

Table 42. Complete Human Review Register.

Review ID	Source Chapter	Question	Classification	Blocking Scope	Upstream Cross-Reference	Status
HR14-1-1	Chapter 1	Confirm NCIE-014 acceptance/sign-off authority (skeletal Preliminary Human	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm NCIE-014 acceptance/sign-off authority (skeletal Preliminary Human	N/A — distinct persistence-engineeri	Open

		Review Theme 1).		Review Theme 1).	ng decision	
HR14-2-1	Chapter 2	Approve persistence principles and unresolved trade-offs (durability/consistency/isolation defaults by store class).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Approve persistence principles and unresolved trade-offs (durability/consistency/isolation defaults by store class).	N/A — distinct persistence-engineering decision	Open
HR14-3-1	Chapter 3	Confirm physical naming/mapping conventions.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm physical naming/mapping conventions.	N/A — distinct persistence-engineering decision	Open
HR14-4-1	Chapter 4	Confirm selected relational technology and transaction requirements (no product mandated by this document per directive item 8).	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)	Confirm selected relational technology and transaction requirements (no product mandated by this document per directive item 8).	N/A — distinct persistence-engineering decision	Open
HR14-5-1	Chapter 5	Confirm analytical-store technology and workload priorities.	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)	Confirm analytical-store technology and workload priorities.	N/A — distinct persistence-engineering decision	Open
HR14-6-1	Chapter 6	Confirm object-store technology and sensitive-object classes.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm object-store technology and sensitive-object classes.	N/A — distinct persistence-engineering decision	Open
HR14-7-1	Chapter 7	Confirm search technology and freshness expectations.	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)	Confirm search technology and freshness expectations.	N/A — distinct persistence-engineering decision	Open
HR14-8-1	Chapter 8	Confirm vector technology, embedding lifecycle and sensitive-data policy.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm vector technology, embedding lifecycle and sensitive-data policy.	N/A — distinct persistence-engineering decision	Open
HR14-9-1	Chapter 9	Confirm graph technology and	ENGINEERING/INFRASTRUCTURE DISCOVERY	Non-blocking — does not gate chapter-level	N/A — distinct	Open

		projection ownership; does not block the reconciliation/rebuild discipline already defined in the interim.	REQUIRED (Non-Blocking)	persistence operation	persiste nce- engineeri ng decision	
HR14-10-1	Chapter 10	Confirm time-series technology and retention/rollup requirements.	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)	Confirm time-series technology and retention/rollup requirements.	N/A — distinct persistence-engineering decision	Open
HR14-11-1	Chapter 11	Confirm physical store choices and history requirements for Agent/Model/Tool/VPF registries.	LEGAL/POLICY CONFIRMATION REQUIRED (Blocking)	Confirm physical store choices and history requirements for Agent/Model/Tool/VPF registries.	N/A — distinct persistence-engineering decision	Open
HR14-12-1	Chapter 12	Confirm store allocation, retention and ephemeral-persistence policy for Memory/Context.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm store allocation, retention and ephemeral-persistence policy for Memory/Context.	N/A — distinct persistence-engineering decision	Open
HR14-13-1	Chapter 13	Confirm storage classes and heightened assurance requirements for Evidence/provenance/audit persistence.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm storage classes and heightened assurance requirements for Evidence/provenance/audit persistence.	N/A — distinct persistence-engineering decision	Open
HR14-14-1	Chapter 14	Confirm which security records may reside in which stores.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm which security records may reside in which stores.	N/A — distinct persistence-engineering decision	Open
HR14-15-1	Chapter 15	Confirm constraint strictness for ingestion/staging versus canonical stores; does not block the baseline constraint discipline already defined in the interim.	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level persistence operation	N/A — distinct persistence-engineering decision	Open
HR14-16-1	Chapter 16	Confirm critical query/workload priorities; does not block the source-supported indexing discipline already	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level persistence operation	N/A — distinct persistence-engineering	Open

		defined in the interim.			decision	
HR14-17-1	Chapter 17	Confirm partition/shard criteria and sovereignty constraints.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm partition/shard criteria and sovereignty constraints.	N/A — distinct persistence-engineering decision	Open
HR14-18-1	Chapter 18	Confirm consistency requirements by workload (acceptable replica lag, authoritative-read routing).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm consistency requirements by workload (acceptable replica lag, authoritative-read routing).	N/A — distinct persistence-engineering decision	Open
HR14-19-1	Chapter 19	Confirm cacheable data classes and invalidation requirements.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm cacheable data classes and invalidation requirements.	N/A — distinct persistence-engineering decision	Open
HR14-20-1	Chapter 20	Confirm approved residency/sovereignty rules and locations covering primary stores, replicas, indexes, caches, backups and archives (Table 24).	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm approved residency/sovereignty rules and locations covering primary stores, replicas, indexes, caches, backups and archives (Table 24).	N/A — distinct persistence-engineering decision	Open
HR14-21-1	Chapter 21	Confirm protection policy by data class.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm protection policy by data class.	N/A — distinct persistence-engineering decision	Open
HR14-22-1	Chapter 22	Confirm retention periods, hold authority and disposal rules (Table 27; no period invented by this document).	LEGAL/POLICY CONFIRMATION REQUIRED (Blocking)	Confirm retention periods, hold authority and disposal rules (Table 27; no period invented by this document).	N/A — distinct persistence-engineering decision	Open
HR14-23-1	Chapter 23	Confirm RPO/RTO and backup classes (Table 29; no numeric value invented by this document).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm RPO/RTO and backup classes (Table 29; no numeric value invented by this document).	N/A — distinct persistence-engineering decision	Open
HR14-24-1	Chapter 24	Confirm DR topology and sovereignty	SECURITY/SOVEREIGNTY CONFIRMATION	Confirm DR topology and sovereignty	N/A — distinct	Open

		constraints.	REQUIRED (Blocking)	constraints.	persiste nce- engineeri ng decision	
HR14 -25-1	Chapt er 25	Confirm migration approval and downtime tolerance.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm migration approval and downtime tolerance.	N/A — distinct persiste nce- engineeri ng decision	Open
HR14 -26-1	Chapt er 26	Confirm legacy sources, acceptance criteria and reconciliation authority.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm legacy sources, acceptance criteria and reconciliation authority.	N/A — distinct persiste nce- engineeri ng decision	Open
HR14 -27-1	Chapt er 27	Confirm auto-repair boundaries and approval.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm auto-repair boundaries and approval.	N/A — distinct persiste nce- engineeri ng decision	Open
HR14 -28-1	Chapt er 28	Confirm DBA privilege model and emergency authority.	SECURITY/SOVEREIGNT Y CONFIRMATION REQUIRED (Blocking)	Confirm DBA privilege model and emergency authority.	N/A — distinct persiste nce- engineeri ng decision	Open
HR14 -29-1	Chapt er 29	Confirm storage SLO/capacity thresholds (no numeric value invented by this document).	ENGINEERING/INFRASTR UCTURE DISCOVERY REQUIRED (Blocking)	Confirm storage SLO/capacity thresholds (no numeric value invented by this document).	N/A — distinct persiste nce- engineeri ng decision	Open
HR14 -30-1	Chapt er 30	Confirm critical workloads and performance targets; does not block the correctness-first optimization principles already defined in the interim.	ENGINEERING/INFRASTR UCTURE DISCOVERY REQUIRED (Non- Blocking)	Non-blocking — does not gate chapter-level persistence operation	N/A — distinct persiste nce- engineeri ng decision	Open
HR14 -31-1	Chapt er 31	Confirm cost objectives and permitted tiering.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm cost objectives and permitted tiering.	N/A — distinct persiste nce- engineeri ng	Open

					decision	
HR14-32-1	Chapter 32	Confirm test acceptance authority and required scenarios.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm test acceptance authority and required scenarios.	N/A — distinct persistence-engineering decision	Open
HR14-33-1	Chapter 33	Review this consolidated register and decide, for each Blocking item, whether it must be resolved before NCIE-014 v1.0 may be approved or may instead remain open under an interim design default.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Review this consolidated register and decide, for each Blocking item, whether it must be resolved before NCIE-014 v1.0 may be approved or may instead remain open under an interim design default.	N/A — distinct persistence-engineering decision	Open
HR14-34-1	Chapter 34	Approve the NCIE-014 skeletal structure and authorize controlled full expansion — already satisfied by the approved NCIE-014 Skeletal v0.1 and the production directive that authorized this v1.0 expansion; preserved here as historical record, not a live open item.	RESOLVED BY UPSTREAM BASELINE (Non-Blocking)	Non-blocking — does not gate chapter-level persistence operation	N/A — distinct persistence-engineering decision	Resolved by upstream baseline

33.4 Mechanical Verification

32 of these 34 items were recomputed programmatically from the Human Review Focus bullet text of Chapters 1-32 rather than assembled by hand. The remaining 2 items are HR14-33-1 (this chapter's own item) and HR14-34-1 (Chapter 34's own item), each defined once and rendered identically in both this table and their own chapter's Human Review Focus bullet. 0 duplicate Review IDs; 0 orphan Human Review items — every Human Review ID appearing anywhere in this specification is reconciled into this register.

33.5 Upstream Deduplication (Directive Item 62)

Every NCIE-014 Human Review item was individually checked against the existing NCIE-008-013 Human Review registers. No NCIE-014 item was found to restate an identical institutional, security, Evidence, integration, UX or domain-business question — each concerns a distinct persistence-engineering decision (technology selection, numeric threshold, store allocation, or residency placement) that those upstream documents raise the need for but do not themselves resolve. No cross-reference deduplication was therefore required for this edition.

33.6 Blocking Scope Discipline

Each Blocking item states exactly what it blocks — a specific technology selection, numeric threshold, retention/residency rule or ownership assignment — rather than implying the whole of NCIE-014 or the whole approved upstream chain is blocked. Non-Blocking items are explicitly proposed defaults or confirmations that do not gate the corresponding chapter's persistence operation in the interim.

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR14-33-1: Review this consolidated register and decide, for each Blocking item, whether it must be resolved before NCIE-014 v1.0 may be approved or may instead remain open under an interim design default.

Traceability: All chapters above (mechanically reconciled register); NCIE-008–013 Human Review registers.

NCIE-014 — Chapter 34 Expansion

Traceability, Acceptance & Downstream Handover

Close traceability to the approved NCIE-001-013 foundation and the approved NCIE-014 Skeletal v0.1; define acceptance criteria and the downstream handover to NCIE-015/016/017.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: Approved NCIE-001-013 foundation.

34.0 Salient Things to Note

Skeleton approval authorizes expansion only; approval does not prove migration, restore, DR, performance or production readiness — §34.4's acceptance gate states this explicitly rather than leaving it implied.

34.1 Traceability Matrix

Every chapter of this document traces to an approved upstream authority or to the approved NCIE-014 Skeletal v0.1 itself. No chapter states a persistence-engineering requirement without an identifiable source.

Table 43. Full traceability matrix.

NCIE-014 Chapter	Primary Upstream Authority	Relationship
Chapter 1	Production-Ready register; NCIE-003 Ch.32; NCIE-004; NCIE-006/009/010/011	Mandate/authority chain/persistence boundary
Chapter 2	NCIE-002 layered architecture; NCIE-003; NCIE-009/010	Persistence engineering doctrine
Chapter 3	NCIE-003 canonical dictionary/ERDs	Canonical-to-physical mapping governance
Chapter 4	NCIE-003 entities; NCIE-004 technology families	Relational/transactional persistence design
Chapter 5	NCIE-002 analytical layer; NCIE-003; NCIE-013	Analytical/columnar persistence design
Chapter 6	NCIE-003 Evidence/source objects; NCIE-010; NCIE-009	Object/blob storage design
Chapter 7	NCIE-006 search; NCIE-010 discovery; NCIE-009	Search index engineering
Chapter 8	NCIE-006 vector/search; NCIE-009; NCIE-010	Vector index/embedding persistence
Chapter 9	NCIE-003 graph/relationships; NCIE-010 provenance graph	Graph persistence & projections
Chapter 10	NCIE-003 temporal/KPI; NCIE-004 observability; NCIE-013	Time-series & metrics persistence
Chapter 11	NCIE-003/004/007; NCIE-009 security	Agent/Model/Tool/VPF registry persistence
Chapter 12	NCIE-006 v1.2; NCIE-009	Memory & Context persistence
Chapter 13	NCIE-010 v1.0; NCIE-009	Evidence, provenance & audit persistence
Chapter 14	NCIE-009 Ch.3-12	Security/identity/secrets/key-metadata persistence interface
Chapter 15	NCIE-003 identifiers/cardinality	Physical keys, constraints & referential integrity

Chapter 16	NCIE-001 workloads; NCIE-003; NCIE-013	Indexing & query access paths
Chapter 17	NCIE-002 scale; NCIE-009 residency; NCIE-003	Partitioning, sharding & distribution
Chapter 18	NCIE-006 current-state; NCIE-009 authz; NCIE-011	Replication, consistency & authoritative-read
Chapter 19	NCIE-006 Context/cache; NCIE-009; NCIE-011	Caching & materialized views
Chapter 20	NCIE-008/009; NCIE-003 classification	Data classification, residency & sovereignty
Chapter 21	NCIE-003 protected identity; NCIE-009 crypto/privacy	Encryption, tokenization & protected fields
Chapter 22	NCIE-006 retention; NCIE-008 governance; NCIE-010 Ch.24	Retention, legal hold, archival & disposition
Chapter 23	NCIE-004 resilience; NCIE-009/010 recovery	Backup, restore & point-in-time recovery
Chapter 24	NCIE-004 DR; NCIE-009; NCIE-010	Disaster recovery & cross-site persistence
Chapter 25	NCIE-003 schema governance; NCIE-004 CI/CD; NCIE-011	Schema migration & change management
Chapter 26	NCIE-003 Ch.34; NCIE-010 provenance	Data migration, backfill & legacy reconciliation
Chapter 27	NCIE-003/010/011	Data lifecycle reconciliation & repair
Chapter 28	NCIE-009 Ch.8-9; NCIE-008 SoD	Database security & privileged administration
Chapter 29	NCIE-004 observability; NCIE-010 distinctions; NCIE-015	Storage observability, capacity & health
Chapter 30	NCIE-002 NFRs; NCIE-013 workloads	Performance engineering & query optimization
Chapter 31	NCIE-004 technology/cost; NCIE-008 governance	Cost, tiering & storage efficiency
Chapter 32	NCIE-016 downstream; NCIE-009/010 assurance	Database/storage testing & verification
Chapter 33	All chapters above; NCIE-008-013 Human Review registers	Consolidated Human Review Register
Chapter 34	Approved NCIE-001-013; NCIE-014 Skeletal v0.1	Traceability/acceptance/handover

34.2 Chapter-Count and Salient-Note Mechanical Verification

A programmatic scan of this document confirms exactly 34 sequential 'NCIE-014 — Chapter N Expansion' labels, N = 1..34, each present exactly once, in order, with no gap, duplicate or unintended renaming (directive item 74). A second scan confirms an explicit '<N>.0 Salient Things to Note' heading present for all 34 chapters (directive item 75), rerun independently against this final document rather than assumed from the build script.

34.3 Skeleton-to-Full Fidelity Verification

Every approved Chapter Purpose, Salient Things to Note, Principal Component, Upstream Dependency/Authority and Human Review Focus from the approved NCIE-014 Skeletal v0.1 is substantively represented in the corresponding full chapter above (directive item 73); none is omitted, and none is conceptually inverted.

34.4 Acceptance Criteria and the Unresolved-Item Gate

Full approval of NCIE-014 v1.0 requires: (a) 34/34 chapter integrity confirmed; (b) 34/34 explicit Salient Things to Note subsections confirmed; (c) skeleton-to-full fidelity confirmed; (d) every Blocking Human Review item in Chapter 33 either resolved by an institutional decision or explicitly and knowingly deferred by the approving authority; (e) the §32.1 test suite specified with no test falsely marked executed, passed, recovery/migration-validated, or accepted; and (f) three-artifact consistency (this document, the Change Register and the Verification Report) confirmed. Approval of the v0.1 skeleton authorized this expansion only — it did not itself approve any technology selection, numeric threshold, retention/residency rule or ownership assignment proposed above. Persistence-architecture approval under this chapter does not prove migration success, restore validity, DR readiness, performance correctness or production readiness.

34.5 Downstream Handover Register

NCIE-014 does not preempt detailed downstream specification. The table below states what is handed over and what remains downstream.

Table 44. Downstream handover register.

Downstream Document	What NCIE-014 Hands Over
NCIE-015 (deployment/operations)	Persistence deployment/infrastructure responsibilities (RACI/control-ownership tables throughout, Chapter 29 observability requirements) for handoff into deployment/operational runbooks; no runbook or infrastructure topology is specified here.
NCIE-016 (detailed testing)	The 9 database/storage test classes specified in §32.1 (Table 39); detailed test cases, execution and recovery/migration validation remain downstream in NCIE-016 — this document specifies the tests, not their execution.
NCIE-017 (implementation roadmap)	The complete set of open persistence-engineering Human Review items (Chapter 33) as build-sequence dependencies; NCIE-014 does not itself sequence implementation work.

34.6 Readiness and Final Disposition

NCIE-014 Database & Storage Engineering Specification v1.0 — Full Expansion of Approved Skeletal v0.1 — is READY FOR HUMAN REVIEW. It is NOT marked Approved. The approved v0.1 skeleton remains historically preserved as the expansion architecture; every proposed technology selection, numeric threshold, retention/residency rule and ownership assignment deferred in Chapters 1-32 remains a proposal pending the Chapter 33 Human Review Register; every §32.1 test remains SPECIFIED, not executed, passed, or recovery/migration-validated, pending NCIE-016.

Human Review Focus

- [RESOLVED BY UPSTREAM BASELINE] (Non-Blocking) HR14-34-1: Approve the NCIE-014 skeletal structure and authorize controlled full expansion — already satisfied by the approved NCIE-014 Skeletal v0.1 and the production directive that authorized this v1.0 expansion; preserved here as historical record, not a live open item.

Traceability: NCIE-001-013 approved foundation; approved NCIE-014 Skeletal v0.1.